Example 6.34

A company buys in lots of 500 boxes which is a 3 month supply. The cost per box is ₹125 and the ordering cost in ₹150. The annual inventory carrying cost is estimated at 20% of unit price.

(i) Determine the total inventory cost of existing inventory policy.

(ii) Determine EOQ in units

(iii) How much money could be saved by applying the economic order quantity?

Solution:

Given

Ordering cost per order: C_3= ₹150 per order.

Number of units per order: q = 500 units

Annual demand = 500 × 4 = 2000 units

∴ Demand rate: R = 2000 per year

Carrying cost: C_1= 20% of unit value

$$C_1 = \frac{20}{100} \times 125 = ₹25$$

(i) Total inventory cost of the existing inventory policy

$$= \frac{R}{q} \times C_3 + \frac{q}{2} C_1 = \frac{2000}{500} \times 150 + \frac{500}{2} \times 25$$

$$= ₹\ 6850$$

(ii) $$EOQ = \sqrt{\frac{2RC_3}{C_1}}$$

$$= \sqrt{\frac{2 \times 2000 \times 150}{25}}$$

$$= \sqrt{12 \times 2000}$$

$$= 155 \text{ units (app.)}$$

(iii) Minimum inventory cost $= \sqrt{2RC_3 C_1}$

$$= \sqrt{2 \times 2000 \times 150 \times 25}$$

$$= ₹3873.$$

By applying the economic order quantity, money saved by a company = 6850–3873

= ₹2977.

Exercise 6.3

1. The following table gives the annual demand and unit price of 3 items

Items	Annual Demand (units)	Unit Price ₹
A	800	0.02
B	400	1.00
C	13,800	0.20

Ordering cost is Rs. 5 per order and annual holding cost is 10% of unit price.

Determine the following:

(i) EOQ in units

(ii) Minimum inventory cost

(iii) EOQ in rupees

(iv) EOQ in years of supply

(v) Number of orders per year.

2. A dealer has to supply his customer with 400 units of a product per every week. The dealer gets the product from the manufacturer at a cost of ₹ 50 per unit. The cost of ordering from the manufacturers in ₹ 75 per order. The cost of holding inventory is 7.5 % per year of the product cost. Find (i) EOQ and (ii) Total optimum cost.

6.4 Partial Derivatives

Partial derivative of a function of several variables is its derivative with respect to one of those variables, keeping other variables as constant. In this section, we will restrict our study to functions of two variables and their derivatives only.

Let $u = f(x, y)$ be a function of two independent variables x and y.

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